RESEARCH REPORT 23 OCTOBER 2026

Ten-variable Diophantine consistency

Four square roots, small rational residues, and a seventeen-integer companion

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AI-assisted mathematical research with literal arithmetic circuits,
fixed integer payloads, ordinary proofs, and independent arithmetic checks.

Abstract

We construct an explicitly specified polynomial in **ten strictly positive integer variables**, of exact degree **1152**, evaluable in **131 binary arithmetic operations**. For the inherited, fixed compact44 coefficient tuple, ordinary mathematics formalizable in ZFC proves that its positive solvability is equivalent to the existence of a finite ZFC refutation. Consequently ZFC proves that its unsolvability is equivalent to Con(ZFC).

The main new step repairs a previously unresolved four-root merger. Instead of assuming that two moduli are coprime, we first recover the Pell indices and then show that the remaining integral rational residue has absolute value less than one. It must vanish. The same estimate rejects the negative main Pell root. Thus one coordinate disappears without imposing any new divisibility condition on the canonical witnesses.

A separate integer construction merges two roots using adjacent-coefficient Pell descent. It gives **seventeen unrestricted integer variables at degree 112**, together with an eighteen-integer degree-80 companion and several intermediate tradeoffs. These are improvements relative to the inspected repository and incoming archives; global minimality and priority in the wider literature are not asserted.

The ten-variable equation retains the native compiler and K215 endpoint, but uses neither the Jones nine-variable compression nor Sun's integer compression. The new reductions are proved here. The inherited native semantic calibration and the outstanding object-language formalization are identified explicitly; this report does not claim a new end-to-end Lean or Rocq certificate.

Reviewed repository snapshot: `c1a102936aca437e9999530a0111e6ddab56dc6d`.
The complete equation, fixed coefficient data, and reproducible checks are included in the accompanying archive.

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1 Outcome, comparison, and logical scope

1.1 The theorem being sought

Fix the recursive syntactic ZFC refutation predicate calibrated by the repository's native source. Its consistency statement is the arithmetical sentence asserting that no finite refutation exists. The target is a fixed integer polynomial P, with a stated witness domain, such that

$$\mathsf{ZFC} \vdash \left[\neg\exists \mathbf{x} \in \mathbb{Z}_{>0}^{10}\ P(\mathbf{x}) = 0\right] \longleftrightarrow \mathrm{Con}(\mathsf{ZFC}). \tag{1}$$

We prove a solvability/refutation equivalence and negate it. Neither consistency nor inconsistency is assumed. A proof of the displayed equivalence does not separately prove either side.

The new argument is an ordinary mathematical proof, using integer arithmetic, finite sequences, Pell recurrences, and finite algebraic extensions. These arguments can be carried out in ZFC. The repository's separate effort to produce an explicit derivation in its encoded ZFC calculus remains a different verification task. The snapshot's `STATUS.md` and `OBJECT_ZFC.md`, retained in the archive, describe that boundary [1].

1.2 Measured results

The following table compares specific constructions, not a hypothetical single ordering of all equations by simplicity. An operation is one live binary addition, subtraction, or multiplication; squaring and multiplication by a fixed scalar are charged. Fixed numeral leaves and reuse are free.

Construction	Domain	Variables	Degree	Operations
Active shared-product baseline	$\mathbb{Z}_{>0}$	9	1,845,940	440
Package 71 lower-degree baseline	$\mathbb{Z}_{>0}$	9	1,379,730	478
Incoming radical-guard reduction	$\mathbb{Z}_{>0}$	9	1,158,228	1753
Incoming centered three-root form	$\mathbb{Z}_{>0}$	11	257	114
Four-root residue equation	$\mathbb{Z}_{>\mathbf{0}}$	**10**	**1152**	**131**
Translated companion	$\mathbb{N}$	10	1152	141
Incoming joint-guard endpoint	$\mathbb{Z}$	19	88	109
Centered pivot companion	$\mathbb{Z}$	**18**	**80**	115
Two-root integer merger	$\mathbb{Z}$	**17**	**112**	124
Sheared integer companion	$\mathbb{Z}$	17	110	140

The ten-positive-variable result occupies a useful gap: one more variable than the nine-variable constructions, but much smaller degree; one fewer variable than the direct eleven-variable construction, at a moderate increase in degree. Its graph contains 77 multiplications, 37 additions, and 17 subtractions. The natural companion replaces every positive coordinate x by $u + 1$, adding ten gates and preserving degree.

The integer companion uses a different fixed tuple and is compared only with integer-domain constructions. Ten unrestricted integers are already available through Sun's compression [11, 7]; we do not claim seventeen is the fewest possible integer variables. The point is the degree/variable tradeoff, without that compression. The supplied integer graphs have measured operation counts; no global circuit minimum is claimed.

1.3 What was actually reviewed

We inspected the placed packages 31–75 and all five incoming archives at the pinned snapshot. Two of the incoming packages already combine signed power recovery, determinant guards, and coordinate projections. Another already proves a four-radical separation lemma but leaves carrier compatibility open. The current report builds on those facts rather than claiming the separation lemma or a previously delivered pivot as new.

The small incoming `report.zip` contains only a main TeX file with missing inputs. Its announced numerical results are distinguished from complete, independently replayable packets in our audit. The four larger archives provide the actual comparison data [6, 7, 8, 9]. Our archive records their Git blob identities and SHA-256 hashes.

2 The complete ten-variable polynomial

2.1 Domain and fixed coefficients

The ten independent variables are exactly

$$j, F, \alpha, i, T, X, s, n, \eta, \zeta \in \mathbb{Z}_{>0}. \tag{2}$$

Every other symbol in this section abbreviates an integer polynomial in these variables. No abbreviation introduces another quantified variable.

The four fixed coefficients are the coherent compact44 tuple

$$(\beta, K_0, M_C, M_F), \qquad \beta = 2^{590069} - 1.$$

The high mask M_F is already shifted by β; it must not be shifted again. The archive contains all four actual integers as compressed, unsigned little-endian byte strings, and reconstructs them from the literal 430-action K215 table using the pinned compact44 recipe. Thus the ports are fixed integer coefficients, not free parameters.

The coefficient bounds used below are

$$K_0 \geq 1, \quad 0 < M_C < \beta, \quad 0 < M_F < 2\beta, \quad \beta \equiv 1 \pmod 3. \tag{3}$$

Their full source meaning is supplied by the compact44 compiler adapter [2]. The arithmetic size of the four payloads is given in Appendix A.

2.2 Outer and Pell registers

Define

$$\begin{aligned}
J &= 3j + 1, & q &= \beta J + 1, & C &= q - F - \alpha,\\
Y &= q^3 s, & E &= XY, & k &= \eta + \zeta,\\
c &= kY + \eta, & a &= E + Y, & H &= 4a + 3,\\
\Delta &= a^2 + H = (a+2)^2 - 1, & \ell &= EkY.
\end{aligned}$$

Next set

$$\begin{aligned}
R &= \big((q-1)(q-F) + \alpha\big)(q^2-1) + (M_C + qM_F)J, & & \\
\widehat{R} &= q^3(q + F + \alpha), & B_t &= (qK_0 + X)C - qF,\\
I &= ic^2, & D_f &= 1 + \Delta I^2,\\
Q &= \Delta^2 I^2, & \delta &= Q - 1,\\
V &= D_f(cT - R) - c, & D_t &= \ell(\ell + k) + 1,\\
D_d &= 1 + \Delta c^2, & \overline{V} &= D_f(cT + \widehat{R}),\\
\overline{\Gamma} &= 2c + 2X + k + \widehat{R} + 1, & L &= 3\overline{V} + \overline{\Gamma},\\
W &= 1 + (q-1)L, & v &= k - R - 1,\\
U &= q^5, & b_0 &= X + ac, \qquad M = (q-1)U.
\end{aligned}$$

The combined carrier is the polynomial

$$Z = M\big[H(\ell + k + n + L + s) + b_0\big] - HUWB_t + H(q-1)(q^4X - v). \tag{4}$$

Its four scaled radicands are represented without division by

$$A = M^2H^2D_t, \quad B = M^2H^2D_f, \quad D = M^2D_d, \quad C_* = M^2H^2(QV^2 - 1). \tag{5}$$

Here C_*/δ will occur only in the proof, not in the equation.

2.3 A short division-free norm

Define the final registers

$$\begin{aligned}
\sigma &= A + B + D, & \pi &= AB + (A + B)D, & \omega &= ABD,\\
a_* &= \delta(Z^2 - \sigma) + C_*, & t_* &= \delta C_* Z^2,\\
k_* &= a_*^2 + 4(t_* - \delta^2\pi), & r_* &= 16\delta^3\omega,\\
b_* &= a_*k_* - r_*,\\
e_* &= k_*^2 + 16t_*a_*^2 - 4r_*(\delta Z^2 + C_*).
\end{aligned}$$

The equation is

$$\boxed{P_{10} = e_*^2 - 64t_*b_*^2 = 0.} \tag{6}$$

All powers are fixed nonnegative integer powers. This is an ordinary polynomial equation with integer coefficients, not an exponential Diophantine equation, a bounded-quantifier abbreviation, or an oracle call. Its literal graph is `ten_positive/generated/p10_positive.json`.

Theorem 2.1. *For the four fixed coefficients specified above,*

$$\exists \mathbf{x} \in \mathbb{Z}_{>0}^{10}\ P_{10}(\mathbf{x}) = 0 \quad \Longleftrightarrow \quad \textit{there is a finite syntactic ZFC refutation.}$$

This equivalence has an ordinary proof formalizable in ZFC. In particular, (1) holds. The polynomial has exact total degree 1152, and the delivered arithmetic graph has 131 live binary operations.

3 What the four-root norm means

For rational arguments define the ordinary algebraic norm polynomial

$$F_{16}(z; a, b, d, c) = \prod_{\varepsilon \in \{-1,1\}^4} (z - \varepsilon_1\sqrt{a} - \varepsilon_2\sqrt{b} - \varepsilon_3\sqrt{d} - \varepsilon_4\sqrt{c}).$$

This symmetric product is a polynomial in z, a, b, d, c with integer coefficients. A useful three-root identity is

$$F_8(z; a, b, d) = \left[(z^2 - a - b - d)^2 - 4(ab + ad + bd)\right]^2 - 64abdz^2. \tag{7}$$

Expanding $F_8(z + \sqrt{c})F_8(z - \sqrt{c})$ into its even and odd parts gives the final schedule in Section 2. More precisely,

$$P_{10} = \delta^8 F_{16}(Z; A, B, D, C_*/\delta). \tag{8}$$

The identity is polynomial after clearing denominators, and is checked by exact arithmetic independently of the graph builder.

At every positive input, $\delta, H, M > 0$, and all four radicands are positive. Hence a zero of P_{10} supplies signs such that

$$\frac{Z}{MH} = \tau_* + f_* + y_* + \frac{d_*}{H}, \qquad \begin{cases} \tau_*^2 = D_t, \\ f_*^2 = D_f, \\ y_*^2 = (QV^2 - 1)/(Q - 1), \\ d_*^2 = D_d. \end{cases} \tag{9}$$

We must prove these roots are integral. Their integrality does not follow merely from taking a norm.

4 Two lemmas for recovering the roots

4.1 Every nonempty subset must be separated

Lemma 4.1. *Let $r_1, \dots, r_m$ be positive real square roots of positive rational numbers. Suppose every nonempty signed subset sum of the r_i is nonzero. If a signed sum of all r_i is rational, every r_i is rational.*

Proof. In the finite multiquadratic field generated by these roots, group the terms according to their square classes in $\mathbb{Q}^*/\mathbb{Q}^{*2}$. Distinct square classes are linearly independent over $\mathbb{Q}$: averaging a proposed relation against the corresponding quadratic characters of the finite Galois group isolates each class. In a rational signed sum, every nontrivial square-class contribution must therefore vanish. A nonempty such class would be a forbidden signed subset sum. All classes are trivial. □

For four roots, distinct radicands and a nonzero total signed sum are insufficient. The identity

$$\sqrt{2} + \sqrt{8} - \sqrt{18} + 100 = 100$$

contains a cancelling triple. One globally dominant root does not repair this defect. The stronger hypothesis in Lemma 4.1 is the one we actually establish.

Let t, d, f, y denote the positive roots of $D_t, D_d, D_f, (QV^2 - 1)/(Q - 1)$. Positivity of the ten inputs alone gives

$$\begin{aligned} &q \geq 16, && Y \geq q^3 \geq 4096, && k \geq 2, && c \geq 2Y + 1, \\ &\ell < t < \ell + k < (a + 1)c, && f > (a + 1)c^2, && 0 < d/H < c. \end{aligned}$$

Since $c < k(Y + 1)$ and $\ell = XkY^2$, we have $t > 2c + 1$. Write $b = cT - R \in \mathbb{Z}$. If $b = 0$, then $V = -c$, so $c < y < c + 1$; therefore the ordering $d/H, y, t, f$ has each term after the first strictly greater than the sum of its predecessors. If $b \neq 0$, then $|V| \geq D_f - c = f^2 - c > 2f$, and $y > |V|$. The ordering $d/H, t, f, y$ has the same property. Here $f - t > (a + 1)c(c - 1)$ and $f > 3c^2$ supply the stated strict gaps.

Thus every nonempty signed subset is separated before any source equation, sign of C, or divisibility conclusion is used. Lemma 4.1 applied to (9) makes $\tau_*, f_*, d_*/H, y_*$ rational. The first three unscaled roots are integers because their squares are integers.

4.2 Adjacent-coefficient Pell descent

Lemma 4.2. *If $Q > 1$ and V are integers, $y \in \mathbb{Q}$, and*

$$QV^2 - (Q - 1)y^2 = 1,$$

then $y \in \mathbb{Z}$.

Proof. Use absolute values and write $y = a/m > 0$ in lowest terms, with $m > 0$. Then $m^2 \mid Q - 1$. Neither coordinate is zero. If $V = 1$, then $y = 1$. For $V > 1$ define

$$V' = (2Q - 1)V - 2(Q - 1)y, \qquad y' = (2Q - 1)y - 2QV.$$

The first coordinate is integral because $m \mid Q - 1$. The norm is unchanged. The inequalities $y > V$ and $QV > (Q - 1)y$ imply $-V < V' < V$, while the norm excludes $V' = 0$. The reduced denominator of y' is still m, since $2Q - 1 \equiv 1 \pmod{m}$. Strong induction on $|V|$ reaches $V = 1$ without changing that denominator; hence $m = 1$. □

This lemma, used in the incoming Pell constructions [4, 6], now gives $y_* \in \mathbb{Z}$. No coprimality of another modulus with $Q - 1$ is needed. Adjacent coefficients matter: $2 \cdot 1^2 - 4(1/2)^2 = 1$ is a counterexample for a general pair of coefficients.

5 Positivity before exact divisibility

The fixed mask bounds imply, for every positive input,

$$|R| < \widehat{R}, \qquad |V| < \overline{V}, \qquad f + y < 3\overline{V}. \tag{10}$$

For the first bound, the mask contribution is less than $(q-1)(1+2q) < 2q^2$, and comparison with $q^3(q+F+\alpha)$ is strict for $q \geq 16, \alpha \geq 1$. The gap $\widehat{R} - |R| \geq 1$, together with $D_f > c$, gives $|V| < D_f(cT + \widehat{R})$. Finally $f < D_f < \overline{V}$ and $y < 2|V| < 2\overline{V}$.

Define two proof-only rational coordinates

$$r = \tau_* - \ell - k - s + \frac{WB_t}{q-1}, \tag{11}$$

$$\gamma = \frac{d_* - b_0}{H} - \frac{q^4X - v}{q^5}. \tag{12}$$

For either sign of d_*,

$$|\gamma| \leq \frac{d}{H} + \frac{b_0}{H} + \frac{X}{q} + \frac{|v|}{q^5} < 2c + 2X + k + \widehat{R} + 1 = \overline{\Gamma}.$$

Equation (9) is exactly

$$r + \gamma + f_* + y_* = n + L. \tag{13}$$

Thus

$$r \geq n + L - f - y - |\gamma| > n > 0.$$

At this stage r and γ may still be rational. The strict inequality is enough. Since $|\tau_*| < \ell + k$,

$$\frac{WB_t}{q-1} = \ell + k + s + r - \tau_* > s + r > 0.$$

Therefore $B_t > 0$, which forces $C > 0$. The authenticated outer packing bounds now give

$$3q < R < q^4 - 2. \tag{14}$$

This order is central: positivity is recovered over the rationals before the two congruence channels are separated over the integers.

6 Recover the Pell indices without choosing the main sign

For $b \geq 2$ define the Pell polynomials by

$$\chi_b(m) + \psi_b(m)\sqrt{b^2-1} = (b + \sqrt{b^2-1})^m.$$

Put $A_0 = a + 2$ and $P_0 = 2XY^2 + 1$. Positive Pell classification gives

$$|d_*| = \chi_{A_0}(p), \quad c = \psi_{A_0}(p), \qquad |\tau_*| = \chi_{P_0}(m), \quad k = 2\psi_{P_0}(m). \tag{15}$$

Only the positive ordinate c and absolute roots occur here. No main-root orientation or positive γ is required.

Since $c > 1$, $p \geq 2$. If $m = 1$, then $k = 2$ and $\eta = \zeta = 1$, giving $c = 2Y + 1 < 2A_0 = \psi_{A_0}(2)$, a contradiction. Hence $m \geq 2$. Since $P_0 > A_0$ and $c > k$, monotonicity gives $m < p$. Moreover

$$c > kY \geq 4P_0Y > 8XY^3 \geq 8q^9 > 2R, \qquad c > 2p. \tag{16}$$

The normalized strong equation is already encoded by D_f:

$$|f_*|^2 - \Delta(ic^2)^2 = 1.$$

Its Pell index m_s satisfies $pc \mid m_s$. Indeed, strong divisibility of the ψ_{A_0} sequence gives $p \mid m_s$. Writing $m_s = pv_s$ and reducing the quotient identity

$$\psi_{A_0}(pv_s)/\psi_{A_0}(p) = \psi_{\chi_{A_0}(p)}(v_s)$$

modulo c shows $c \mid v_s$, since $\chi_{A_0}(p)^2 \equiv 1 \pmod{c}$. In particular $c \mid m_s$ and $m_s > 2p$.

For clarity, the inherited signed auxiliary-index lemma used next has the following hypotheses: $c = \psi_{A_0}(p)$, $f = \chi_{A_0}(m_s)$, $c \mid m_s$, $m_s > 2p$; $S \geq 2$, $S^2 = \Delta(f^2 - 1)$, and $c \mid S$; $S^2V^2 - (S^2 - 1)y^2 = 1$; $V \equiv -c \pmod{f}$, $V \equiv -R \pmod{c}$; and $0 < p, R < c/2$. It permits signed V, y and does not assume any main-root congruence [3, lemmas on odd subsequences, signed index recovery, and normalized rank].

Here take $S = \Delta I$. Then $S^2 = Q$, while $V = D_f(cT - R) - c$ has both required congruences because $f^2 \equiv 1 \pmod{c}$. All interval hypotheses follow from (16). The lemma gives

$$p = R. \tag{17}$$

Its proof uses the odd Pell identities to obtain $R \equiv \pm p \pmod{c}$, followed by the strict half-interval bounds; the full inherited argument is included in the archive.

7 Separate the two residues by size

7.1 The q-power channel

All starred roots are now integral, so (13) implies $r + \gamma \in \mathbb{Z}$. Clearing the denominators in (11)–(12) gives

$$q^5 \mid H(q-1)(q^4X - v).$$

Since $q \equiv 2 \pmod 3$, $q^3 \mid Y$, and $H = 4Y(X+1)+3$, we have $\gcd(H, q) = 1$. Also $\gcd(q, q-1) = 1$. Consequently

$$q^5 \mid q^4X - v, \qquad q^4 \mid v. \tag{18}$$

The first Pell recurrence gives $k \equiv 2m \pmod{4XY^2}$, whose modulus is divisible by q^4. Thus $2m - R - 1 \equiv 0 \pmod{q^4}$. Since $1 \leq m < p = R < q^4$, that difference has absolute value less than q^4. It is zero:

$$2m = R + 1. \tag{19}$$

Now $v = k - 2m$ is a positive multiple of $4XY^2$, hence of q^5. Equation (18) therefore forces $q \mid X$. Returning to the merged equality, whose q^5 fraction is now integral, gives

$$\frac{(q-1)(d_* - X - ac)}{H} \in \mathbb{Z}. \tag{20}$$

We have not cancelled $q - 1$ modulo H.

7.2 The bound which makes cancellation unnecessary

Put $r_0 = (R-1)/2 = m - 1 \geq 24$. Recurrence induction gives

$$\psi_b(t) > (2b-1)^{t-1} \quad (t \geq 2), \qquad \psi_b(t) \leq (2b)^{t-1} \quad (t \geq 1).$$

The exact identity

$$(2A_0 - 1)^2 - 2P_0X = 4Y^2(2X+1) + 12Y(X+1) + 9 - 2X > 0$$

yields

$$\frac{c}{k} = \frac{\psi_{A_0}(2r_0+1)}{2\psi_{P_0}(r_0+1)} > \frac{X^{r_0}}{2}.$$

But $c/k = Y + \eta/k < Y + 1$, so, using $X \geq q \geq 16$,

$$Y > \frac{X^{r_0}}{2} - 1 \geq \frac{q^{r_0}}{2} - 1 > q\,2^{2r_0+1} = q\,2^R. \tag{21}$$

The last elementary inequality already holds for $r_0 \geq 4$.

7.3 Reject the negative main root and recover the power

The Pell polynomial identities modulo H are

$$\chi_{A_0}(R) - a\psi_{A_0}(R) \equiv 2^R \pmod{H}, \tag{22}$$

$$2^R\big(\chi_{A_0}(R) + a\psi_{A_0}(R)\big) \equiv 1 \pmod{H}. \tag{23}$$

They follow by substituting $\Delta \equiv a^2 \pmod{H}$ into the Pell binomial expressions: $A_0 - a = 2$, $A_0 + a = 2a + 2$, and $2(2a + 2) = H + 1$.

If $d_* = -\chi_{A_0}(R)$, then (20) and (23) imply

$$\frac{(q-1)(1+2^R X)}{H} \in \mathbb{Z}.$$

This number is positive but, by (21), is strictly less than one. That is impossible. Hence $d_* > 0$.

For the positive root, (20) and (22) give

$$\frac{(q-1)(2^R - X)}{H} \in \mathbb{Z}.$$

Its absolute value is less than one because

$$H > 4q\,2^R(X+1) > q(2^R + X) > (q-1)|2^R - X|.$$

It must be zero. We have proved

$$X = 2^R, \qquad H \mid d_* - X - ac. \tag{24}$$

The proof-only γ and then r are integral. Since $W \equiv 1 \pmod{q-1}$, their defining identities also give

$$q - 1 \mid B_t. \tag{25}$$

This is the new residue-separation argument. It uses an exact integer interval, with no new condition on $\gcd(H, q-1)$ or on canonical Y.

8 Soundness and completeness of the new equation

8.1 Reconstruct a positive normalized source solution

Define

$$w = X/q, \quad h = v/(XY), \quad \gamma_{\mathrm{old}} = (d_* - X - ac)/H, \quad f_{\mathrm{old}} = |f_*|, \quad y_{\mathrm{old}} = |y_*|.$$

These are integers. They are positive: $w > 0$; $h > 0$ follows from $v = k - 2m > 0$; and $d_* > (a+1)c$, $c > X$ give $H\gamma_{\mathrm{old}} > c - X > 0$. The first, main, strong, and auxiliary Pell equations now hold, as does $k - hXY - R = 1$.

Because $B_t = q[(K_0 + w)C - F] > 0$, (25) gives the positive integer

$$z_{\mathrm{new}} = \frac{(K_0 + w)C - F}{q-1}.$$

Set $z = z_{\mathrm{new}} + 1$. Then the source transport identity is exactly

$$(K_0 + w)C + (q - F) - z(q-1) = 1.$$

Finally set $T_{\mathrm{old}} = f_{\mathrm{old}} T$. The source auxiliary coordinate agrees because

$$c(T_{\mathrm{old}} f_{\mathrm{old}} - 1) - R f_{\mathrm{old}}^2 = D_f(cT - R) - c = V.$$

Thus we have a positive solution of the normalized compact44 source from package 61. Its compiler/calibration theorem gives a finite ZFC refutation. No search for a refutation is performed by the verification programs.

8.2 Construct a new zero from a canonical source solution

Conversely a finite refutation gives canonical source witnesses. The source already has $Y = q^3 s$ with $s > 0$; no additional congruence on s is imposed. Its padded repunit $J = 1 + B + \cdots + B^{N-1}$ has odd $N > 1$ and $B = 2^{590069} \equiv 2 \pmod 3$. Hence $J \equiv 1 \pmod 3$ and $j = (J-1)/3 > 0$.

Choose the normalized strong pair $f, I = ic^2$ and the positive auxiliary pair as in the source, using its canonical $R \equiv 3 \pmod 4$ condition. Compute L, W after these choices. The source gives $t_0 = B_t/(q-1) \geq q$, $\ell < \tau < \ell + k$, and $s < k$. Set

$$r = Wt_0 + \tau - \ell - k - s, \qquad \gamma = \gamma_{\mathrm{old}} - X/q + v/q^5.$$

Both are integers; v is a positive multiple of $4XY^2$ and $q \mid X$. The unconditional bound $|\gamma| < \overline{\Gamma} < L$ applies. Also $L > k$, so

$$r > qW - 2k > 2L.$$

Now take

$$n = r + \gamma + f + y - L > r - |\gamma| - L > 0.$$

The four-root equality holds with the positive roots, and therefore $P_{10} = 0$. All ten coordinates are strictly positive. This completes both directions of Theorem 2.1, subject exactly to the stated inherited source theorem.

9 Exact degree and arithmetic cost

Every input in (2) has degree one; the four fixed ports have degree zero. The relevant register degrees are

	Y	c	Δ	I	D_f	Q	V	L	Z	A	B	C_*
deg	4	5	10	11	32	42	38	38	51	42	54	140

and $\deg D = 32$. In the last norm schedule,

$$\deg a_* = 144, \quad \deg k_* = 288, \quad \deg b_* = 432, \quad \deg e_* = 576.$$

The subtracted term $64 t_* b_*^2$ has degree 1148, below the degree 1152 of e_*^2. Thus the leading form is $Q_0^8 Z_0^{16}$.

To prove that the degree remains exact on the actual coefficient slice, put

$$q_0 = 3\beta j, \quad k_0 = \eta + \zeta, \quad Y_0 = q_0^3 s, \quad B_{t,0} = (K_0 q_0 + X)(q_0 - F - \alpha) - q_0 F.$$

Then

$$\begin{aligned} Q_0 &= i^2 k_0^4 X^4 Y_0^8, \\ Z_0 &= -12 q_0^6 i^2 k_0^5 X^3 Y_0^8 T B_{t,0}, \\ [P_{10}]_{1152} &= 12^{16} q_0^{672} i^{48} k_0^{112} X^{80} s^{192} T^{16} B_{t,0}^{16}. \end{aligned}$$

In particular the coefficient of

$$j^{672} i^{48} \eta^{112} X^{96} s^{192} T^{16} \alpha^{16} \tag{26}$$

is exactly

$$12^{16} (3\beta)^{672} > 0.$$

The minimum j-degree and the absence of F in the target monomial force $-X\alpha$ from each factor of $B_{t,0}$; no specialization of K_0, M_C, M_F can cancel this coefficient. The monomial has degree 1152, proving exactness rather than merely a syntactic upper bound.

The graph builder shares q^2, q^3, q^4, $D_f - 1$, kY, and the repeated norm products. The independent checker traverses the live graph, checks the complete declared input set, and recounts every operation. The result is $77 + 37 + 17 = 131$. These figures do not measure bit complexity: the fixed coefficients and potential witnesses are large.

10 The unrestricted-integer companions

The following separate reduction starts from the incoming joint-guard integer family [6, 7]. It uses the source34 coefficient tuple, together with two fixed cone scales. It does not silently substitute the compact44 tuple used in P_{10}.

10.1 A transport pivot beyond the earlier projections

The retained transport equation becomes monic in F after using the complement coordinate C:

$$\alpha = q - F - C, \qquad F = (K_0 + W)C - z(q-1).$$

These are polynomial substitutions over $\mathbb{Z}$. Forgetting the eliminated coordinate and restoring it by this formula are inverse on the integer zero sets. Applying this previously known pivot to the more deeply projected joint-guard packets gives the following new combinations:

integer variables	21	20	19	18
exact degree	16	20	32	88

The pivot raises the packing expression's degree only to five and the determinant guard's degree only to four, below the existing dominant rows. The delivered packets include their complete restoration maps and exact degree certificates.

10.2 Centering one Pell root without a parity witness

For the first Pell residual write $u = E(c - \eta)$. Instead of the old root τ_{old}, set $t = 2\tau_{\mathrm{old}} - 2u - k$. Its residual is

$$t^2 + 2(2u+k)t + k^2 - 4 = 0. \tag{27}$$

Modulo four this forces $t + k$ even, so $\tau_{\mathrm{old}} = (t + 2u + k)/2$ is an integer automatically. Equation (27) is exactly four times the old residual under this substitution. The high-degree u^2 term disappears.

In the eighteen-variable packet this lowers the first residual's degree from 34 to 18. Let G be the remaining joint positivity residual of degree 44, and let S be the sum of squares of the other residuals, of degree 36. Since $S \geq 0$ over $\mathbb{Z}$,

$$(1+G)(1+S) - 1 = 0 \quad \Longleftrightarrow \quad G = 0 \text{ and } S = 0.$$

Indeed $1 + S$ is a positive integer divisor of one. The resulting single polynomial has exact degree $44 + 36 = 80$ in eighteen integers.

10.3 A second root merger gives seventeen integers

Retain the strong equation and put

$$I = (2r+1)c^2, \quad Q = I^2, \quad A = u(u+k) + 1, \quad B = \frac{QV^2 - 1}{Q - 1}.$$

The old first and auxiliary roots satisfy $\tau^2 = A$ and $y^2 = B$. Replace their two coordinates by the single integer

$$\xi = 2(I\tau + y) - I(2u + k).$$

Write $s_I = (\xi + I(2u+k))/2$ in the proof only. The denominator-cleared two-root norm is

$$\big[(Q-1)(4s_I^2 - 4QA) - 4(QV^2 - 1)\big]^2 - 64(Q-1)QA(QV^2 - 1) = 0. \tag{28}$$

Here $4s_I^2 = (\xi + I(2u+k))^2$, so the displayed equation is polynomial in the retained integer variables. The literal packet uses the equivalent centered expression to avoid unnecessary high-degree cancellation.

The retained joint guard proves the nondegeneracy needed for this merger: $|c| > 1$, $Q > 1$, and $A > 0$, before the deleted root conditions are used. The two rational radicands QA and B cannot coincide. Otherwise

$$(Q-1)QA = QV^2 - 1$$

would contradict reduction modulo Q. If $B < 0$, the norm cannot have a real zero; the case $B = 0$ is also impossible modulo Q. For positive, distinct radicands, a rational signed sum of the two roots makes both roots rational. Therefore τ is integral by rational-square integrality, and y is integral by Lemma 4.2. This recovers the two old rows without an extra parity or divisibility witness.

Conversely, old integer roots give the displayed integer ξ. Thus the root replacement preserves existential solvability in both directions. Together with the earlier transport pivot this gives a single sum of squares in **seventeen unrestricted integers**, of exact degree **112**. The archive includes all residuals, their variable list, coefficient binding, restoration maps, and independent checks. Its supplied graph uses 124 operations, without a claim of operation minimality.

10.4 Complete display of the seventeen-variable equation

For an explicit standalone specification, the seventeen unrestricted integer coordinates are

$$J_3, C, z, f, h, r, v, w_3, \gamma, o, j, c, t, p_1, p_2, p_3, \xi.$$

Here z remains the transport coordinate and t is the joint-guard root coordinate. Set $\lambda = 2^{1113}$ and $\mu = 2^{666104}$; the other four fixed coefficients are the complete source34 tuple β, K_0, M_C, M_F, with $\beta = 2^{1064921} - 1$ and M_F already shifted. Use only the following polynomial abbreviations:

$$\begin{aligned}
J &= 3J_3 + 1, & q &= \beta J + 1, & W &= 4(3w_3 + 1),\\
X &= qW, & Y &= (2v+1)q^3, & E &= XY,\\
a &= (X+1)Y, & H &= 4a + 3, & \Delta &= a^2 + H,\\
V &= fo - c, & k &= 1 - V + jc + hE, & \eta &= c - Yk,\\
u &= EYk, & A &= u(u+k) + 1, & I &= (2r+1)c^2,\\
Q &= I^2, & \delta &= Q - 1, & F &= (K_0 + W)C - z(q-1).
\end{aligned}$$

Further put

$$\begin{aligned}
R &= \big(q(q-F) - C\big)(q^2-1) + (M_C + qM_F)J,\\
d &= (\lambda F)(\mu C) - (q - \lambda F - \mu C)^2,\\
T &= \eta(k - \eta),\\
P_{\mathrm{pos}} &= 1 + p_1^2 + p_2^2 + p_3(p_3+1),\\
G &= dT - (t^2 - d - T + 1)^2 - P_{\mathrm{pos}},\\
B_0 &= QV^2 - 1,\\
L &= \xi^2 + 2I(2u+k)\xi + Q(k^2-4),\\
D_n &= \delta L - 4B_0,\\
N &= D_n^2 - 64\delta QAB_0.
\end{aligned}$$

The promised single equation is

$$\boxed{P_{17} = M^2 + B_p^2 + S_p^2 + G^2 + N^2 = 0,} \tag{29}$$

where

$$M = (X + ac + \gamma H)^2 - \Delta c^2 - 1, \quad B_p = cj - V - R, \quad S_p = Q - \Delta(f^2 - 1).$$

There are no hidden quantified coordinates or divisions in this display. Its five residual degrees are $14, 5, 14, 44, 56$. The coefficient of

$$r^{12}c^{24}h^4f^4o^4J_3^{44}w_3^8v^{12}$$

in P_{17} is exactly $2^{52}3^{52}\beta^{44} > 0$, certifying degree 112 after specialization to the actual coefficient tuple. The supplied literal arithmetic DAG uses 124 binary operations under the stated counting convention; this is a verified construction, with no claim of operation minimality.

The scale I in the merger is essential. Taking $I = 2$, $Q = 4$, $V = 2$, and $A = 5$ makes the unscaled auxiliary radicand $(QV^2 - 1)/(Q - 1) = 5$. The unscaled norm then accepts the spurious rational sum $\sqrt{5} - \sqrt{5} = 0$, although neither square root is integral. The scaled radicands are 20 and 5, and the same sum-zero input is rejected.

10.5 A degree-110 integer companion

A polynomial change of coordinate $\xi_{\text{old}} = \xi + 2V$ gives a separate seventeen-integer equation of degree 110. Keep the definition of L above, now using the new coordinate ξ, and set

$$\begin{aligned} J_* &= \delta(L + 4V\xi) - 4(V^2 - 1), \\ H_* &= 4\delta I(2u + k)V, \\ N_{110} &= J_*^2 + 2J_*H_* + 64\delta QA(1 - V^2) + 16\delta^2 Q(k^2 - 4)V^2. \end{aligned}$$

The exact identity $N_{110}(\xi) = N(\xi + 2V)$ follows from $(2u + k)^2 = 4A + k^2 - 4$ and $\delta = Q - 1$. Replace N^2 by N_{110}^2 in (29); the other four rows are unchanged. The coordinate change is invertible over the integers, so solvability is preserved. The coefficient of

$$r^{12}c^{24}h^4f^2o^2\xi^2J_3^{44}w_3^8v^{12}$$

is $2^{52}3^{52}\beta^{44} > 0$, proving exact degree 110. The supplied circuit uses 140 operations. The degree-112 version remains the principal integer display because its circuit is shorter, at 124 operations; neither choice dominates the other on both measures.

11 Why the new argument is different

The new ten-variable merger has three logically separate stages. First, a strict hierarchy of all four root magnitudes makes a rational signed sum force rationality of every root. Second, a sufficiently large positive transport margin establishes $C > 0$ while its recovered quotient is still only rational. Third, Pell growth makes the remaining integral rational residue too small to be nonzero.

This last stage removes the earlier obstruction. From $H \mid (q - 1)(d - X - ac)$ one generally cannot cancel $q - 1$. Nor can one freely impose $(q - 1) \mid Y$: Y is tied to a half-binomial value, not an arbitrary large slack. The new proof requires neither move.

There is a concrete control even within the exact half-binomial kernel. Take $q = 32$, $R = 655351$, and $r_0 = (R - 1)/2 = 327675$. Then $\text{popcount}(r_0) = 16$, and for the kernel's exact half-binomial ordinate Y one has

$$v_2(Y) = 15, \quad X = 2^R \equiv 2 \pmod{31}, \quad Y \equiv 23 \pmod{31}, \quad H \equiv 0 \pmod{31}.$$

Thus $q^3 \mid Y$ but $\gcd(H, q - 1) = 31$. The modular residue is checked by a finite Lucas-digit computation, cross-checked against direct binomial sums on small inputs. This is a kernel example, not a zero for the authentic fixed outer tuple. Its role is to show why the proof must not assume universal coprimality.

The integer companion uses a related but simpler idea: normalize the auxiliary equation to adjacent coefficients before merging its root. The descent lemma restores integrality that a bare resultant would not guarantee. Both improvements change the dependency order of the proof, not merely the naming of an existing arithmetic register.

12 Reproducibility, evidence, and remaining work

12.1 What is supplied and checked

The archive contains the complete report source and PDF, the literal positive and natural arithmetic graphs, the integer residual packets, all required coefficient payloads, deterministic builders, independent checkers, upstream proof notes, source pins, and attribution notices.

The ten-variable verification has distinct layers:

1. Reconstruct the entire coefficient tuple from the literal machine, verify its finite geometry and padding certificates, and compare every integer byte with the bound payloads.
2. Verify the four-root norm by exact polynomial arithmetic, independently of the circuit builder. No random evaluation stands in for that identity.
3. Check the graph's topology, liveness, full input set, exact schedule, operation count, degree bound, and nonzero leading certificate.
4. Compare independently implemented evaluations using the actual fixed coefficients modulo primes, as transcription diagnostics.
5. Exercise controls for radical cancellation, nonadjacent Pell descent, and the failure of a coprimality assumption. Mutating a critical norm coefficient must be rejected.

The integer component separately checks monic restoration, centered-root parity, two-root elimination, exact degrees, and its fixed coefficient bindings. The root runner records each component's result.

12.2 What finite verification cannot establish

Finite tests do not prove the unbounded Pell or compiler theorems. Those are mathematical arguments with named source dependencies. The report's new arithmetic proof has received an independent adversarial review, but it is not a kernel-checked Lean or Rocq theorem. The native compiler's object-language ZFC internalization remains open in the reviewed snapshot.

The construction is explicit in the standard computational sense: a finite graph and fully specified integer coefficients. Expanding every enormous coefficient and every monomial into a single line would not simplify the mathematics. The complete fixed integer bytes and rebuilding recipes are included rather than replaced by an unspecified existence claim.

12.3 Useful next steps

The immediate verification target is the new residue-separation theorem, with its sign recovery and canonical completeness, then the literal four-root graph. The theorem is modular: it requires the same normalized source interface already isolated in the repository.

A later size search can center the new carrier or reduce its majorant, provided the rational positivity bootstrap survives. Another possible direction is to adapt the same small-residue principle to different proof-certificate encodings. Neither proposed extension is counted as a completed equation in this report. The delivered results are the concrete ten-positive-variable and integer companion constructions above.

A Fixed coefficients and exact archive definitions

For P_{10}, each listed gzip payload decompresses to a byte string $b_0, \ldots, b_{m-1}$ and denotes exactly $\sum_{j=0}^{m-1} b_j 256^j$. There is no sign byte or truncation.

Port	Bits	Raw bytes	Payload
β	590069	73759	`beta.bin.gz`
K_0	592884	74111	`K.bin.gz`
M_C	590069	73759	`MC.bin.gz`
M_F	590070	73759	`MF_source.bin.gz`
Total		295388	

The full hashes are in `ten_positive/coefficients/BINDING.json`. The unchanged compact44 generator is retained with its attribution and its literal transition data. `rebuild_coefficients.py` reconstructs all four values in a temporary directory and compares the complete raw integer bytes. The native radix is $\rho = 2^{67}$ with cell length 8807; therefore $B = \rho^{8807} = 2^{590069}$. The compiler uses the literal 943-tile language, complement fields, duplicated clauses, and its specified padding controller [2]. All those finite constants are fixed by the included recipe.

The integer companion's six coefficient ports are separately stored and bound under `integer/coefficients/`. They are based on $\beta = 2^{1064921} - 1$, with fixed cone scales 2^{1113} and 2^{666104}. The compact44 payloads must not be substituted into that construction.

B Files and commands

From the archive root, run

```
python3 verify_all.py
```

The main ten-variable component uses only the Python standard library. The integer symbolic builder has its pinned requirements in its own `requirements.txt`; the archive's README explains installation when needed. To rebuild the report, run

```
pdflatex -interaction=nonstopmode -halt-on-error report.tex
pdflatex -interaction=nonstopmode -halt-on-error report.tex
```

No script needs a GitHub connection to define or evaluate the delivered polynomials. All proof-source paths are pinned in source manifests.

References

[1] V. Reshetnikov, *ProveIt: DiophantineConsistency active research*, snapshot `c1a102936aca437e9999530a0111e6ddab56dc6d`, especially `README.md`, `SIZE_REDUCTION.md`, `STATUS.md`, and `OBJECT_ZFC.md`. https://github.com/VladimirReshetnikov/ProveIt/tree/c1a102936aca437e9999530a0111e6ddab56dc6d/Computability/DiophantineConsistency.

[2] ProveIt package 44, *Complement-packed Wang943 coefficients*, `44-b4-sparse/native/PROOF.md`, coefficient recipe, and bindings. Pinned sources and actual bytes retained in this archive.

[3] ProveIt package 61, *Congruence carriers*, `report/congruence_carriers.tex`, `positive/PROOF13.md`, and `positive/PROOF14.md`. The normalized source interface, direct strong-rank lemma, signed auxiliary-index lemma, and canonical witness construction are used explicitly.

[4] ProveIt package 69, *Weighted cones and rational Pell descent*, especially `positive/PROOF.md`.

[5] ProveIt package 70, *Residue orientation and root transfer*, especially the eleven-positive-variable proof and signed power-recovery theorem.

[6] *ZFC Diophantine Joint Guards*, incoming archive `ZFC_Diophantine_Joint_Guards_2026-10-24.zip`, at the reviewed snapshot. Used components: `pell/`, `joint/`, and `circuits/`. The archive's UTC date differs from this report's Washington State date.

[7] *Structural Reductions, Round 11*, incoming archive `ConZFC_Structural_Reductions_Round11_2026-10-24.zip`, especially the transport pivot and calibrated cone proof.

[8] *Radical guard descent*, incoming archive `con-zfc-radical-guard-descent-2026-10-23.zip`; used for the current comparison frontier.

[9] *Budget and centering*, incoming archive `con-zfc-budget-and-centering-2026-10-23.zip`; in particular its four-radical separation discussion and carrier-compatibility boundary.

[10] ProveIt native stream queue, *A strong half-binomial Pell kernel in 42 operations*, `pell_kernel_half_binomial42.md`. The retained note supplies the exact half-binomial ordinate and valuation formula used only in the coprimality control.

[11] Z.-W. Sun, *Ten unknowns for Hilbert's tenth problem over the integers*, arXiv:2609.01594v2 (2026). https://arxiv.org/abs/2609.01594.